

Weekly Notice 6

Exercises, week 6: In several exercises, we need to find a stationary distribution. For this purpose, we can often use Theorem 6.23, where (6.12) provides a representation of the stationary distribution. It is worth noting that we can choose the value of j freely, which can be extremely useful in many cases. However, in order to use (6.12), it is required that we can calculate γ^j , which is certainly not always the case. Examples where γ^j (with an appropriate choice of j) is known explicitly are presented in Exercises 54 and 57 – and in the Mandatory Assignment. At other times, however, one must solve $\pi = \pi P$ by using brute force ¹.

Sometimes one has a Markov chain with a positive recurrent part and some transient states, and in that case, Exercise 63 can be useful.

1. Exercise 54. The last question uses the Law of Large Numbers - which is part of the Monday lecture.
2. Consider the transition matrix in Exercise 58(2). Does there exist a stationary distribution? If so, is it unique?
3. Exercise 47. In part (2) only prove that B_1 and B_2 are independent under $\mathbb{P}_i$, and that $\mathbb{P}_i(B_1) = \mathbb{P}_i(B_2)$.
4. Exercise 63.

¹and, if one is really lucky, the creator of the exercise has specified π , so you just need to check this equality.